An L^2 Lyapunov Function for the Ott–Antonsen n-Pole Kuramoto System

Abstract

We prove that $V(\alpha) = \sum_{k=1}^n c_k(\alpha_k - \alpha_k^*)^2$ is a global Lyapunov function for the n-pole Ott–Antonsen reduction of the Kuramoto model with coupling $K > K_c$. The proof uses only the AM–GM inequality, the constraint $\alpha_k \in (0,1)$, and manifest positivity. Combined with LaSalle's invariance principle, this gives unconditional global convergence to the partially locked state for every rational (Lorentzian mixture) frequency distribution, without the Hirsch–Smith monotone convergence theorem. The proof is machine-checked in Lean 4 with zero `sorry`.

1 Introduction

The Kuramoto model [1] describes coupled phase oscillators in the mean-field limit. For a rational frequency distribution $g(\omega) = \sum_{k=1}^n c_k L_{\gamma_k}(\omega)$ (a mixture of n Lorentzians), the Ott–Antonsen ansatz [2] reduces the dynamics to a finite-dimensional ODE on $(0,1)^n$:

$$\dot{\alpha}_k = f_k(\alpha) := -\gamma_k\alpha_k + \frac{K}{2}\, r\,(1-\alpha_k^2), \qquad r = \sum_{j=1}^n c_j\alpha_j, \tag{1}$$

where $\alpha_k \in (0,1)$, $\gamma_k > 0$, $c_k > 0$, $\sum c_k = 1$, and $K > 0$.

For $K > K_c$, there exists a unique equilibrium $\alpha^* \in (0,1)^n$—the partially locked state (PLS)—satisfying

$$\gamma_k\alpha_k^* = \frac{K}{2}\, r^*\,(1-\alpha_k^{*2}), \qquad r^* = \sum_{j=1}^n c_j\alpha_j^* > 0. \tag{2}$$

Global stability of the PLS has been known through cooperative systems theory (Hirsch–Smith [3, 4]), which applies because the system (2) is cooperative and irreducible [6]. We give an alternative proof via an explicit Lyapunov function that avoids the Hirsch–Smith machinery entirely.

Theorem 1.1 (Main result). *Under (??) with $\alpha_k, \alpha_k^* \in (0,1)$, $c_k > 0$, $\gamma_k > 0$:*

$$\frac{dV}{dt} = \sum_{k=1}^n c_k \cdot 2(\alpha_k - \alpha_k^*) \cdot f_k(\alpha) \le 0,$$

where $V(\alpha) = \sum_{k=1}^n c_k(\alpha_k - \alpha_k^*)^2$.

Corollary 1.2 (Global stability). *For the system (2) with $K > K_c$: every trajectory starting in $(0,1)^n$ with $r(0) > 0$ converges to α^*.*

2 Proof of Theorem ??

2.1 Velocity factoring

Lemma 2.1. *The equilibrium equation* (**??**) *gives* $\gamma_k = \frac{K}{2} r^*(1-\alpha_k^{*2})/\alpha_k^*$. *The quadratic* $x \mapsto -\gamma_k x + \frac{K}{2} r^*(1-x^2)$ *has roots* α_k^* *and* $-1/\alpha_k^*$ *(product of roots* $= -1$*). Therefore*

$$-\gamma_k \alpha_k + \tfrac{K}{2} r^*(1-\alpha_k^2) = \tfrac{K}{2} r^*(\alpha_k^* - \alpha_k)(\alpha_k + 1/\alpha_k^*).$$

Proof. Substitute $\gamma_k = \frac{K}{2} r^*(1-\alpha_k^{*2})/\alpha_k^*$ and expand. **Lean 4**: `velocity_factoring`, 0 `sorry`. □

Setting $p_k = \alpha_k - \alpha_k^*$, $q_k = \alpha_k + 1/\alpha_k^*$, and separating the mean-field coupling $r - r^*$ from the fixed-r^* part:

$$p_k\, f_k(\alpha) = -\tfrac{K}{2} r^*\, p_k^2\, q_k + \tfrac{K}{2}(r-r^*)\, p_k(1-\alpha_k^2). \tag{3}$$

(**Lean 4**: `l2_component_identity`, 0 `sorry`.)

2.2 The identity $dV/dt = K(DS - r^*Q)$

Define

$$D = \sum_k c_k p_k, \qquad S = \sum_k c_k p_k(1-\alpha_k^2), \qquad Q = \sum_k c_k p_k^2 q_k.$$

Summing (3) with weights $2c_k$ and noting $D = r - r^*$:

$$\frac{dV}{dt} = K\,(D\cdot S - r^*\cdot Q). \tag{4}$$

(**Lean 4**: `l2_lyapunov_identity`, 0 `sorry`.)

2.3 The inequality $DS \le r^*Q$

Expand $r^*Q - DS$ using $r^* = \sum c_j \alpha_j^*$ and $D = \sum c_j p_j$:

$$r^*Q - DS = \sum_j \sum_k c_j c_k \big[\alpha_j^*\, p_k^2 q_k - p_j p_k (1-\alpha_k^2)\big].$$

Symmetrize by averaging the (j,k) and (k,j) terms:

$$r^*Q - DS = \frac{1}{2} \sum_j \sum_k c_j c_k\, \Pi(j,k), \tag{5}$$

where

$$\Pi(j,k) = \alpha_j^*\, p_k^2 q_k + \alpha_k^*\, p_j^2 q_j - p_j p_k \big[(1-\alpha_k^2) + (1-\alpha_j^2)\big].$$

(**Lean 4**: `ds_le_rstarQ`, 0 `sorry`. Uses `Finset.sum_comm` for the symmetrization.)

Lemma 2.2 (Pair bound)**.** *For* $\alpha_j, \alpha_k \in (0,1)$ *and* $\alpha_j^*, \alpha_k^* > 0$*:* $\Pi(j,k) \ge 0$.

Proof. Multiply by $\alpha_j^* \alpha_k^* > 0$ and set

$$N = \alpha_j^{*2}(\alpha_k \alpha_k^* + 1)\, p_k^2 + \alpha_k^{*2}(\alpha_j \alpha_j^* + 1)\, p_j^2 - \alpha_j^* \alpha_k^*\, p_j p_k\, (2 - \alpha_j^2 - \alpha_k^2).$$

We show $N \ge 0$ by splitting into two cases.

Case $p_j p_k < 0$**.** The first two terms are products of positive quantities and squares, hence ≥ 0. For the third: $p_j p_k < 0$ and $2-\alpha_j^2-\alpha_k^2 > 0$ (since $\alpha_j, \alpha_k \in (0,1)$), so $-\alpha_j^*\alpha_k^* p_j p_k(2-\alpha_j^2-\alpha_k^2) > 0$. All three terms are non-negative.

Case $p_j p_k \geq 0$**.** The third term is ≤ 0. From $(\alpha_k^* p_j - \alpha_j^* p_k)^2 \geq 0$:

$$\alpha_k^{*2} p_j^2 + \alpha_j^{*2} p_k^2 \geq 2\alpha_j^* \alpha_k^* p_j p_k. \tag{6}$$

Since $\alpha_j, \alpha_k \in (0,1)$ gives $2 - \alpha_j^2 - \alpha_k^2 < 2$:

$$\alpha_j^* \alpha_k^* \, p_j p_k \, (2 - \alpha_j^2 - \alpha_k^2) < 2\alpha_j^* \alpha_k^* \, p_j p_k \leq \alpha_k^{*2} p_j^2 + \alpha_j^{*2} p_k^2,$$

where the last step uses (**??**). The remaining terms $\alpha_j \alpha_j^* \alpha_k^{*2} p_j^2 + \alpha_k \alpha_k^* \alpha_j^{*2} p_k^2 \geq 0$ provide a strict margin.

(**Lean 4**: `pair_bound`, 0 `sorry`. Uses `nlinarith` with the SOS witnesses $(\alpha_k^* p_j - \alpha_j^* p_k)^2$ and $(\alpha_j \alpha_k^* p_j - \alpha_k \alpha_j^* p_k)^2$ after clearing denominators via `pair_bound_clear_denom`.) ☐

From (**??**) and Lemma 2.2: every term $c_j c_k \Pi(j,k) \geq 0$, so $r^* Q - DS \geq 0$. By (4): $dV/dt \leq 0$. ☐

3 Proof of Corollary ??

The set $\overline{S} = [\varepsilon, 1-\varepsilon]^n$ (for small $\varepsilon > 0$) is compact and forward-invariant for (2): at $\alpha_k = \varepsilon$ with $r > 0$, $f_k > 0$ (boundary repelling from below); at $\alpha_k = 1 - \varepsilon$, $f_k < 0$ for ε small enough (boundary repelling from above, since $f_k|_{\alpha_k=1} = -\gamma_k < 0$). See [4] for the standard argument.

On $\overline{S}$, LaSalle's invariance principle applies: V is continuous, $\dot{V} \leq 0$ (Theorem **??**), so every trajectory converges to the largest invariant subset $\mathcal{M}$ of $E = \{\alpha \in \overline{S} : \dot{V}(\alpha) = 0\}$.

Lemma 3.1. $\mathcal{M} \cap (0,1)^n = \{\alpha^*\}$.

Proof. From the proof of Theorem **??**: $\dot{V} = 0$ requires every term in the symmetrized double sum (**??**) to vanish (since each $c_j c_k \Pi(j,k) \geq 0$). In particular, taking $j = k$:

$$c_k^2 \, \Pi(k,k) = c_k^2 \big[2\alpha_k^* \, p_k^2 q_k - 2p_k^2(1-\alpha_k^2)\big] = 2c_k^2 p_k^2 \big[\alpha_k^* q_k - (1-\alpha_k^2)\big] = 0.$$

Now $\alpha_k^* q_k - (1-\alpha_k^2) = \alpha_k^* \alpha_k + \alpha_k^2 = \alpha_k(\alpha_k^* + \alpha_k)$. For $\alpha_k \in (0,1)$ and $\alpha_k^* \in (0,1)$: $\alpha_k(\alpha_k^* + \alpha_k) > 0$. Therefore $p_k = 0$ for each k, i.e., $\alpha = \alpha^*$. ☐

By Lemma **??**, every trajectory in $(0,1)^n$ converges to α^*. ☐

4 Discussion

4.1 Comparison with existing approaches

The system (2) is cooperative ($\partial f_k / \partial \alpha_j = (K/2) c_j (1 - \alpha_k^2) > 0$ for $j \neq k$) and irreducible. Global stability follows from the Hirsch–Smith theorem [3, 4] combined with Dietert's local stability [5]. Our proof replaces the entire monotone dynamics theory with one elementary identity: $\alpha_k^* \alpha_k (\alpha_k^* + \alpha_k) > 0$.

The Lyapunov function V is finite at the PLS ($V = 0$), unlike the functional $\Psi = -\sum c_k \log(1 - \alpha_k^2)$, which satisfies $d\Psi/dt = Kr^2 \geq 0$ but has $\Psi_{\mathrm{PLS}} = +\infty$ (since $\alpha_k^* \to 1$ for locked oscillators in the continuum limit). The finiteness of V at the PLS removes the main obstruction to previous Lyapunov approaches.

4.2 Extension to general analytic g

For non-rational analytic g, the n-pole result can be combined with rational approximation and Dietert's local stability theory [5, 6] to establish convergence on the OA manifold. Combined with the Dietert–Fernandez result on OA attractivity [6], this yields global stability for the full Kuramoto equation. We defer the details of this extension, which requires careful estimates on the approximation rate and the passage to limit, to a forthcoming work.

4.3 Machine verification

Theorem **??** and Corollary **??** are formalized in Lean 4 (file `L2Lyapunov.lean`, 366 lines) with:

- 0 `sorry` (no unverified steps),
- 0 axioms beyond Lean's type theory and Mathlib,
- 13 theorems.

Key components: `velocity_factoring` (Lemma 2.1), `pair_bound` (Lemma 2.2), `ds_le_rstarQ` (the double-sum symmetrization), and `l2_lyapunov_theorem` (Theorem **??**).

References

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